



Fig. 3: Author's prototype of transmitter module placed in a toy car

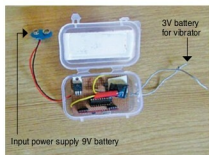


Fig. 4: Author's prototype of receiver module enclosed in a box

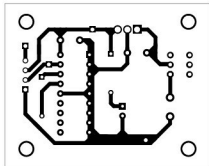


Fig. 5: Actual-size PCB pattern for the transmitter circuit

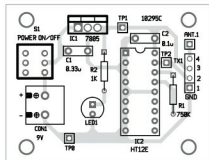


Fig. 6: Component layout for the PCB in Fig. 5

433MHz frequency ASK technique for communication. Their maximum range is 100 metres when used with a 17cm long antenna. In this ap-

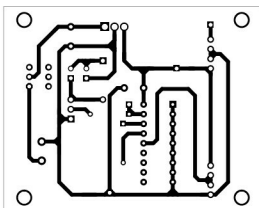


Fig. 7: Actual-size PCB pattern for the receiver circuit

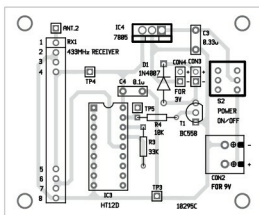


Fig. 8: Component layout for the PCB in Fig. 7

Test Points

Test point	Details
TP0, TP3	0V (GND)
TP1, TP4	5V when switches S1 and S2 are on
TP2	Pulses
TP5	High during transmission

plication, if we need only one-metre range the antennae may not be required at all.

In normal mode, as soon as the modules are supplied with power, both transmitter and receiver get activated. Then pin 17 of IC3 goes high, indicating that the connection between the transmitter and the receiver has been successfully established. If there is a failure of transmission, pin 17 of IC3 remains low.

As pin 17 of IC3 is connected to pnp transistor BC558 we get complement output. The transmitter circuit

may be hidden in the child's clothes or in a toy/box attached to the child while the receiver remains with the accompanying elder person.

When the child moves away from the parent, after the set range, connection breaks between the transmitter and receiver modules and pin 17 of decoder goes low. The pnp transistor conducts and the vibrator starts vibrating. This alerts the elder that the child is moving away beyond the secure range.

Construction and testing

The transmitter may be connected to a 9V battery and placed in a toy car as shown in Fig. 3. The receiver module may be put in a box as shown in Fig. 4. (An external antenna could be used for increasing the range, if so desired.) It is better to use a separate 3V supply for the receiver circuit and vibrator, otherwise it may not work.

An actual-size, single-side PCB for the transmitter part is shown in Fig. 5 and its component layout in Fig. 6. Actual-size, single-side PCB for the receiver part is shown in Fig. 7 and its component layout in Fig. 8. After assembling the circuits on respective PCBs, enclose these in two separate plastic boxes.

The transmitter as well as the receiver circuit works off a 9V battery or regulated power supply. For troubleshooting, if needed, you may refer to the test points table for checking the voltages that may be expected at various points of the circuit. **EFY**



Sachin K. is a final-year diploma student in E&C, SDM Polytechnic, Ujire